INVESTIGATING THE STABILITY OF SPINNING WORMHOLES WITH NONLINEAR EVOLUTIONS

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We investigate the stability of spinning Ellis–Bronnikov wormholes using fully nonlinear 3+1 numerical relativity simulations of the drainhole family, whose static member we evolved in our previous work. We construct the stationary metrics with a spectral solver that is exact to $\sim 10^{-10}$ at slow spin and $\sim 10^{-7}$ near the extremal limit, and validate it against the published symmetric family of Kleihaus and Kunz. In doing so we correct an inconsistency in a published field equation and map the drainhole family, for which no fast-spin solutions were known, up to $v_e = 0.99$; the 3D initial data converge at fourth order. We find that rotation slows the radial instability and then changes its fate. Along the ladder of evolved spins the collapse e-fold time grows from 4.4 (static) to 8.0 at 24% of the extremal angular momentum; at 29% and 33% the unperturbed throat turns and inflates instead of collapsing—the instability's other branch, with the seed's sign selecting the fate—and at 81% the throat is stable to $t = 150$ at three resolutions. The measured rate tracks the second-order prediction of Azad *et al.* where both exist and extrapolates to zero at ≈42% of the extremal angular momentum. The unperturbed rotating collapse radiates an $\ell = 2$, $m = 0$ burst whose energy grows with spin, while the static collapse radiates nothing; second-order perturbative initial data instead seed the instability at the level of their own truncation error.

I. INTRODUCTION

The Ellis–Bronnikov wormhole, a throat held open by a phantom scalar field [1–4], is unstable: depending on the sign of the perturbation it collapses to a black hole or inflates, on a timescale set by the throat radius [5–9]. Nonlinear 3+1 evolutions confirm both branches for the lone throat [10] and carry them into binaries, whose every channel ends in collapse, inflation or a merged horizon within a few dynamical times [11]. If traversable wormholes are to persist anywhere, something must shut this instability off.

Rotation is the oldest candidate [12]. Spinning Ellis–Bronnikov solutions are known perturbatively [13] and, beyond perturbation theory, from the elliptic solver of Kleihaus and Kunz [14, 15]: a family that starts at the static throat and ends, at equatorial rotation velocity $v_e = 1$, on an extremal Kerr black hole, the phantom field vanishing in the limit; no closed form exists [16]. On the stability side, Azad, Blázquez-Salcedo, Khoo and Kunz [17, 18] carried the radial mode to second order in spin: the static unstable mode weakens, with a positive $O(J^2)$ correction for every asymmetry, while for nearly symmetric wormholes a second unstable mode grows out of a static zero mode; the two cross near 30% of the limiting angular momentum, where the quadratic approximation gives out, and the authors conjecture the modes merge and the instability disappears, as purely imaginary modes do in five dimensions [19]. The charged analogue warns against optimism: there the merged modes continue as a complex pair, growth that oscillates, and the growth rate reaches zero only in the extremal limit [20]. Which continuation rotation takes in four dimensions is exactly the question a perturbative method cannot answer, and no rotating wormhole has ever been evolved nonlinearly.

This paper builds the tools for that evolution and takes its first step. We construct two families of backgrounds as exact stationary solutions. The symmetric family ($\gamma = 0$), massless when static, is the one with published data—the curves of Ref. [14] and the scalar charges of Ref. [15]—and we use it to validate the solver and the 3D initial data across the spin range. The drainhole family ($\gamma = -\pi$), whose static member is the mass-1 drainhole evolved in Refs. [10, 11], has no published fast-spin data at all; it is the family we evolve. The solver is gated on identities it never imposes, a discipline that pays immediately: one of the published field equations is inconsistent (Sec. II A), and second-order perturbative initial data turn out to seed the instability at their own error level (Sec. V). We then put the backgrounds on the AMR grid, verify them at fourth order, and evolve the drainhole along a ladder of nine spins (Table III). The result: rotation slows the collapse, as the perturbative sign predicts, and near 30% of the extremal angular momentum the same mode changes fate—the unperturbed throat turns and inflates—while at 81% the throat is stable over the full run.

Section II gives the family and the corrected field equations; Sec. III the solver, the validation against the published data, and the map of the parameter space (Table II); Sec. IV the numerical setup and the runs (Table III); Sec. V the evolutions; Sec. VI the conclusions. Appendix A holds solver details; Appendix B the convergence of the evolutions at the two ends of the ladder. We use $G = c = 1$. Table II quotes the backgrounds in

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units of the throat parameter η_0; the evolutions use the code units of Refs. [10, 11], $\eta_0 = 2$, in which the static drainhole has mass $M = 1$.

II. THE SPINNING WORMHOLE FAMILY

The matter model is that of our previous work [10, 11]: Einstein gravity coupled to a phantom (ghost) scalar field ψ,

$$S = \frac{1}{16\pi}\int d^4x\,\sqrt{-g}\,(R + 2\,g^{\mu\nu}\partial_\mu\psi\,\partial_\nu\psi)\,, \qquad (1)$$

whose kinetic term has the reversed sign that makes ψ a ghost. Varying the action with respect to the metric and to ψ gives the field equations of the model [Eqs. (2)–(3) of Ref. [15]],

$$R_{\mu\nu} = -2\,\partial_\mu\psi\,\partial_\nu\psi, \qquad \partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\psi\big) = 0\,: \qquad (2)$$

Einstein's equations with the phantom's stress tensor as source, and the wave equation for the phantom on the curved background. Those papers evolved its best-known static wormhole—the Ellis–Bronnikov drainhole with throat parameter $a = 2$ and end masses ±1—and measured the instability that this paper asks rotation to cure. Spinning that wormhole up requires the stationary rotating solution of the same theory, and none is known in closed form [16]: it has to be constructed numerically, inside an *ansatz*—an assumed form of the metric with stationarity and axisymmetry built in and a few unknown functions left for the field equations to determine. We take the ansatz of Refs. [14, 15], who first constructed these solutions,

$$\begin{aligned} ds^2 = &-e^f dt^2 + p^2 e^{-f}\Big[e^\nu\left(d\eta^2 + h\,d\theta^2\right)\\ &+ h\sin^2\theta\,(d\varphi - \omega\,dt)^2\Big], \end{aligned} \qquad (3)$$

with $\eta \in (-\infty, \infty)$ and four unknown functions f, p, ν, ω of η and θ [Eq. (4) of Ref. [15]]. The field equation for p decouples from the others, and its solution with asymptotically flat boundary conditions is $p = 1$ [Eqs. (5)–(6) of Ref. [15]]; we set $p = 1$ from here on. The fixed function $h = \eta^2 + \eta_0^2$ is what makes this a wormhole ansatz. In the trivial limit $f = \nu = \omega = 0$ the spatial metric is $d\eta^2 + h\left(d\theta^2 + \sin^2\theta\,d\varphi^2\right)$, so $\sqrt{h}$ is the radius of the sphere at coordinate η—and by construction it never vanishes: it falls from η^2 at large $|\eta|$, bottoms out at η_0 at $\eta = 0$, and rises again. The spacetime therefore has no center; a minimal sphere of radius set by η_0, the throat, joins the two asymptotically flat ends $\eta \to \pm\infty$. Choosing h hard-wires this two-ended, horizonless topology into Eq. (3), while f, ν, ω deform the geometry on top of it (h also recurs as the weight in every equation below); conversely, $\eta_0 \to 0$ closes the throat and lets the same ansatz hold a black hole, which is how extremal Kerr enters in Sec. II A. Our previous wormhole sits inside this form: with $\omega = \nu = 0$ and the closed-form arctan profile for f [15], Eq. (3) is the static drainhole of Refs. [10, 11]—η is the isotropic radius r used there, unfolded through the throat by $\eta = r - \eta_0^2/4r$ (Sec. IV), and $\eta_0 = a$ its throat parameter—while $f = 0$ gives the massless Ellis wormhole. Everything the ansatz (3) leaves open—the metric functions $f(\eta,\theta)$ (the redshift of the two ends), $\omega(\eta,\theta)$ (the frame dragging) and $\nu(\eta,\theta)$, together with the phantom field—must come from the field equations (2): determining them at each spin is what Sec. III is for. Substituting the ansatz (3) into the field equations (2) reduces them in three steps. We follow the reduction of Kleihaus and Kunz [14], given in full in Sec. II of Ref. [15]; equation numbers in brackets below refer to Ref. [15]. First, the phantom field equation—the wave equation in Eq. (2), which stationarity and axisymmetry reduce to a function of η alone—integrates in closed form to $\psi = (D/\eta_0)\,[\arctan(\eta/\eta_0) - \pi/2]$ [Eqs. (7)–(9)]: the scalar enters the rest of the problem only through its constant of integration D, the scalar charge. Second, because $\partial_\mu\psi$ points along η, the source term $-2\,\partial_\mu\psi\,\partial_\nu\psi$ survives only in the $\eta\eta$ Einstein equation; each other nontrivial component (tt, $t\varphi$, $\theta\theta$, $\varphi\varphi$, $\eta\theta$) is a statement about the metric functions alone, and these organize by order and role. Two are second-order elliptic equations,

$$\begin{aligned} 0 = &\,\partial_\eta(hs\,\partial_\eta f) + \partial_\theta(s\,\partial_\theta f)\\ &- hs^3 e^{-2f}\left[h(\partial_\eta\omega)^2 + (\partial_\theta\omega)^2\right], \end{aligned} \qquad (4)$$

$$\begin{aligned} 0 = &\,\partial_\eta\left(h^2 s^3 e^{-2f}\partial_\eta\omega\right)\\ &+ \partial_\theta\left(hs^3 e^{-2f}\partial_\theta\omega\right), \end{aligned} \qquad (5)$$

with $s \equiv \sin\theta$ [Eqs. (10)–(11) of Ref. [15]]: Eq. (4) is a Laplace-type equation for the redshift f, sourced by the squared gradients of the frame dragging—rotation gravitates—and Eq. (5) is a total divergence, the local form of angular-momentum conservation, which integrated over the domain forces the same J through every $\eta = $ const surface. The single off-diagonal component is only first order in ν,

$$\begin{aligned} 0 = &-hs\,\partial_\eta f\,\partial_\theta f + h\cos\theta\,\partial_\eta\nu\\ &+ \eta s\,\partial_\theta\nu + h^2 s^3 e^{-2f}\partial_\eta\omega\,\partial_\theta\omega, \end{aligned} \qquad (6)$$

so ν follows by integration once f and ω are known [Eq. (13)]. (The remaining diagonal components give a second-order equation for ν as well—the published Eq. (12), shown misprinted in Sec. II A; we never use it.) Third, the one sourced component, $R_{\eta\eta} = -2D^2/h^2$ [Eq. (18)], rearranges into an algebraic identity for the charge [Eq. (19)]: the metric functions fix D^2 at every point, so its constancy over the domain is a solution test that comes for free. Three unknown functions thus face more equations than they need; the Bianchi identities make the overdetermined set consistent on a true solution, and a numerical scheme must choose which subset to impose—a choice Sec. II A shows matters. The boundary conditions select the family: Minkowski at $\eta \to +\infty$, and $f \to \gamma$, $\omega \to \omega_\infty$ at $\eta \to -\infty$, so the constant γ

measures how unequal the two ends are and ω_∞ sets the spin.

A solution of this system is three functions and one number, $(f, \nu, \omega; D)$. Every physical quantity we quote—the columns of Table II—is read off them directly [14, 15]:

- far from the throat the metric settles to the field of a point mass, $f \to -2\mu_+/\eta$ as $\eta \to +\infty$ and $f \to \gamma + 2\mu_-/\eta$ as $\eta \to -\infty$, so the end masses $\mu_\pm$ are the $1/\eta$ coefficients of f;
- likewise $\omega \to 2J/\eta^3$, and the conservation law (5) forces the same J on both ends; the charges obey the Smarr-type relation $\mu_+ + \mu_- = 2\,\omega_\infty J$ [15], which the solver never imposes and uses as an acceptance gate;
- the throat is the sphere of minimal area, at $\eta_t = 0$ when the two ends are equal ($\gamma = 0$) and displaced toward the positive-mass end otherwise ($\eta_t = \eta_0/2$ for the static drainhole): its equatorial circumference gives the radius $R_e = e^{-f/2}\sqrt{h}\,\big|_{\eta_t,\,\theta=\pi/2}$, its meridians the polar radius R_p, its area the areal radius R_A—three numbers that coincide while the throat is round and split as rotation flattens it;
- the throat is dragged at the angular velocity $\Omega = \omega\,\big|_{\eta_t,\,\theta=\pi/2}$, and $v_e = R_e\,\Omega$ is its equatorial linear velocity in units of c: the dimensionless spin label used throughout, 0 when static and 1 at the extremal Kerr endpoint.

The same read-offs, applied to the 3D grid data instead of the solver's functions, are how Sec. IV verifies the initial data against Table II.

We construct two one-parameter families, labelled by ω_∞ (equivalently J or v_e):

(a) *the drainhole family*, $\eta_0 = 2$, $\gamma = -\pi$: its static member is the Ellis drainhole with throat parameter $a = 2$ and unequal end masses $\mu_+ = 1$, $\mu_- = -1$, the wormhole of Refs. [10, 11] (the asymmetry $C = 1$ of Refs. [17, 18]). This is the family we evolve.

(b) *the symmetric family*, $\gamma = 0$ ($C = 0$): equal masses on both ends, massless and identical to the Ellis wormhole when static. This is the family of Refs. [14, 15], the only one with published solutions, and we use it to validate the solver (Sec. III) and the 3D initial data (Sec. IV); for the 3D checks it is rescaled $\eta_0 = 1 \to 2$, so its throat sits on family (a)'s grid.

A. An inconsistency in the published field equations

Refs. [14, 15] close the system with a second-order equation for ν, Eq. (12) of Ref. [15]. That equation is inconsistent with the rest. Extremal Kerr fits the ansatz (3) in closed form (with $\eta_0 = 0$), and substituted symbolically, at 160-digit arithmetic, it satisfies Eqs. (4)–(6) and the D^2 identity exactly—while leaving an $O(10^{-2})$ residue in the printed ν equation alone. The error also shows dynamically: solutions of the printed elliptic system converge at every resolution, yet carry a violation of the unimposed Eq. (6) and D^2 identity that scales as J^4 and does not shrink with the grid—the fingerprint of a wrong continuous equation, not of numerics (first seen as a spread of 5.5×10^{-7} in D^2 at $J = 0.25$). We therefore never solve the printed ν equation: ν and D^2 are obtained jointly from Eq. (6) and the D^2 identity, and the printed equation is kept only as a reported diagnostic. Every published conclusion of Refs. [14, 15] that we can test survives the correction (Sec. III, Appendix A); the published positive-asymmetry scalar charges do not (Table I).

III. BACKGROUNDS ACROSS THE SPIN RANGE

The solver turns the system of Sec. II into numbers in five steps (Fig. 1); Appendix A gives the details. First, the infinite domain is made finite: the coordinate $x = (2/\pi)\arctan(\eta/\eta_0)$ maps both asymptotic ends to $x = \pm 1$, and the unknowns f and ω are expanded in Chebyshev polynomials of x and even cosine modes of θ. The expansion is spectral: for smooth solutions the error falls exponentially with the number of modes, so modest grids reach near machine precision. Second, the two elliptic equations (4)–(5) are solved for the expansion coefficients by Newton iteration—guess, linearize, correct—until their residual sits at round-off. Third, ν and the scalar charge are recovered without ever touching the misprinted second-order ν equation: the first-order constraint (6) and the D^2 identity together overdetermine ν and the single number D^2, so both are extracted as one linear least-squares problem—and the redundancy is a feature, since the leftover misfit measures the error. Fourth, each family is walked in spin: every converged solution seeds the Newton iteration at a slightly higher spin, with J as the continuation parameter (ω_∞ is not monotone along a family; Appendix A), and the grid refines automatically, from 48×12 up to 240×60, whenever the acceptance gates fail. Fifth—the point of the whole design—a solution is accepted only on identities the solver never imposes: the spread of D^2 over the domain ($\le 10^{-11}$), the Smarr relation ($\le 10^{-10}$), equality of the angular-momentum flux through the two ends ($\le 1.3 \times 10^{-9}$), the asymptotic tails ($\le 5 \times 10^{-12}$), and an independent, symbolically generated residual of the full Einstein equations, evaluated out to 20 throat radii: 1.0×10^{-10} at $J = 0.25$ and 4.2×10^{-10} at $J = 1$ against the static member's 2×10^{-11} round-off floor—where the $O(J^2)$ perturbative data give 3×10^{-5} and 1×10^{-2} on the same meshes—staying near 10^{-9} up to $v_e \approx 0.85$ and reaching $\sim 10^{-7}$ at $v_e = 0.91$–0.99. The high-spin 10^{-7} is

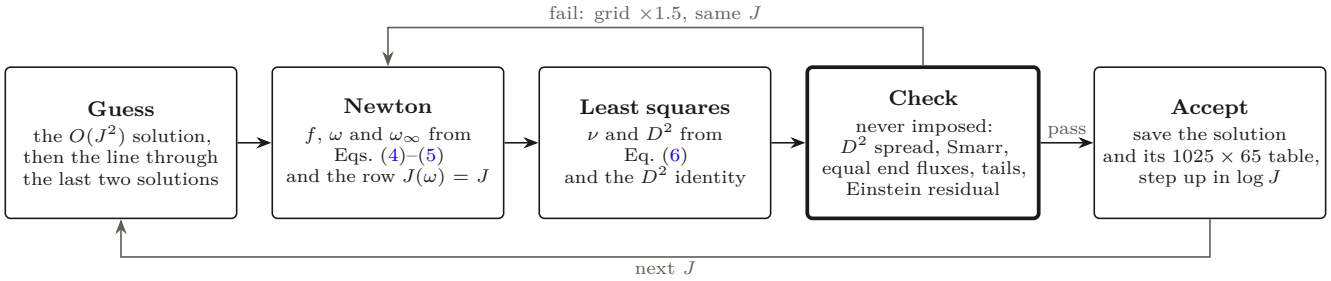


FIG. 1. The solver loop for one family (Sec. III). Thin boxes impose equations; the heavy box tests identities the solver never imposes, and only it decides acceptance. A failed check refines the Chebyshev grid (48×12 up to 240×60) and solves the same J again; an accepted solution seeds the next J. The loop gave 462 accepted solutions over 20 continuation legs, both families to $v_e \approx 0.99$.

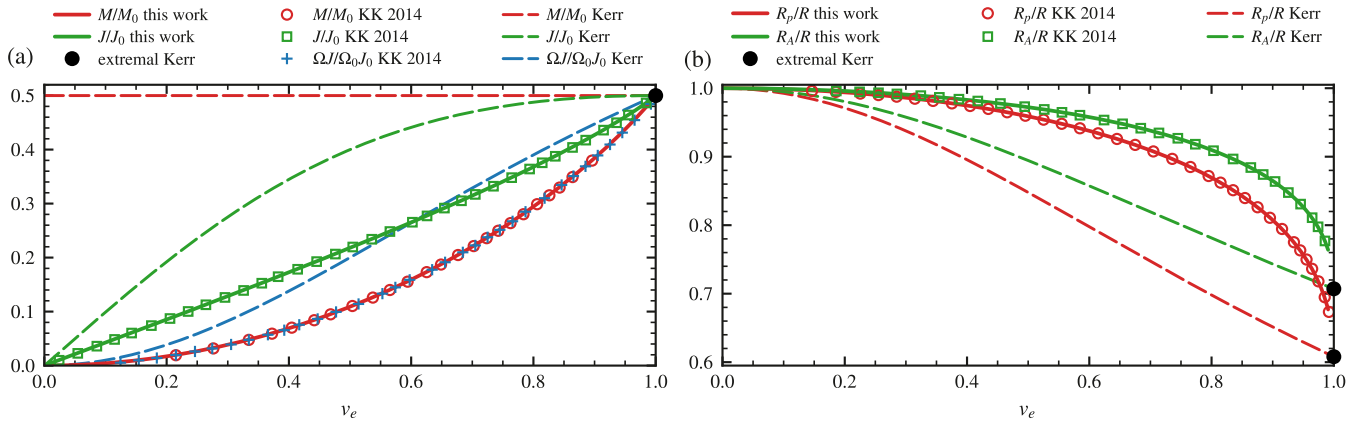


FIG. 2. The symmetric family ($\eta_0 = 1$, $\gamma = 0$; 81 solutions, $v_e = 0.02$–0.991, plus the exact static member) in the normalization of Ref. [14] ($M_0 = R_e$, $J_0 = R_e^2/2$, $\Omega_0 = 1/R_e$): our curves, the digitized points of their Fig. 1 (open symbols), and closed-form Kerr (dashed). The digitization's reading error, measured by reading their printed Kerr curves the same way, is rms 1–6×10^{-4}; digitized minus ours is rms 2.5–8×10^{-4}, max 2.5×10^{-3}, over all five curves. The two solvers agree to the precision a printed figure carries.

round-off, not solution error: it does not shrink on finer grids, while the mass and scalar charge come out identical to 12 digits across them; and the test does catch genuine failures, scoring a deliberately under-resolved solution at 10^{-4}, which the resolution check rejects. Beyond 20 throat radii the residual's normalization inflates round-off and the test is switched off; that outer region is covered by the resolution study, by the end charges, and by the 3D constraint checks of Sec. IV. The products: 462 converged solutions over 20 continuation legs, global charges stable to 10 digits across three grids, and, for evolution, each background exported as a 1025×65 table whose bilinear interpolation error is $\leq 10^{-7}$ up to $v_e = 0.9$.

Figure 2 is the validation: in the normalization of Ref. [14], our symmetric family lands on the digitized curves of their Fig. 1 to the digitization's own reading error. Their four anchor solutions at $\omega_\infty = 0.5$ [15] split: the $\gamma = 0$ and $\gamma = -1$ scalar charges agree to all printed digits, while $\gamma = 1$ and $\gamma = 2$ differ by 2×10^{-4} and 6.6×10^{-3} from our grid-converged values—their finite-difference error, growing with γ (Table I).

TABLE I. Scalar charge D of the anchor solutions ($\eta_0 = 1$, $\omega_\infty = 0.5$) against Ref. [15]. Along each fixed-γ family, ω_∞ rises, turns over and descends, so $\omega_\infty = 0.5$ is reached twice; the published values sit on the post-turnover branch. Our values are grid-converged to the digits shown; the $\gamma = 2$ family continued to $J = 172$ confirms no third crossing exists. The two positive-γ offsets are the published values' finite-difference error, growing with γ; the pre-turnover branch is new.

γ	D (this work)	D [15]	D (pre-turnover, new)
-1	0.716368	0.71636	0.969194
0	0.552771	0.55277	0.986013
1	0.4314654	0.43155	1.008136
2	0.3375993	0.33984	1.048161

Figure 3 and Table II are the map of the parameter space the foundation papers left open. Published data exist only for the symmetric family and only as figures; the

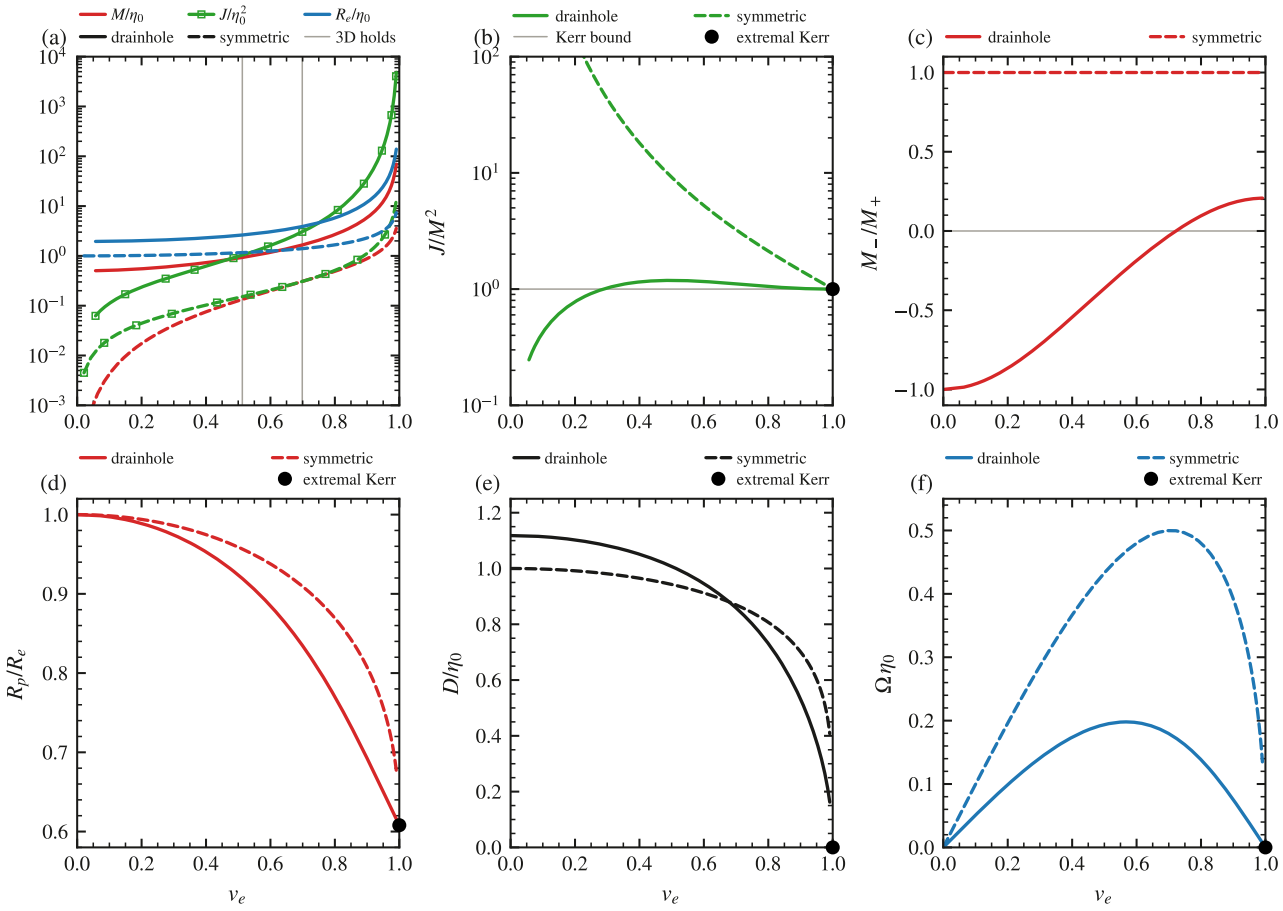


FIG. 3. The two families across the spin range (solid: drainhole, $\eta_0 = 2$, $\gamma = -\pi$; dashed: symmetric, $\eta_0 = 1$, $\gamma = 0$), against v_e: (a) mass, angular momentum and equatorial throat radius; (b) J/M^2 against the Kerr bound; (c) the ratio of end masses; (d) the flattening R_p/R_e; (e) the scalar charge; (f) the throat angular velocity, which peaks and falls to zero at extremality. Grey verticals in (a): the 0.2c and 0.5c rungs of the evolution ladder (Table III). The symmetric family violates the Kerr bound at every spin and the drainhole family exceeds it by up to $\sim 20\%$ at intermediate spin; both families end on extremal Kerr.

massive drainhole family—the one whose static member has been evolved, and the one from which binaries can be built [11]—is mapped here from slow spin to $v_e = 0.99$. Two of its features matter later. First, both families violate the Kerr bound: the symmetric family everywhere ($J/\mu_+^2 \approx 85$ already at $v_e = 0.24$, falling monotonically to 1 at extremality, since its static member is massless), the drainhole family by up to $\sim 20\%$ at intermediate spin. A collapsing overspun throat cannot settle to a Kerr black hole of the same mass and spin, which makes these backgrounds sharp cosmic-censorship initial data. Second, the drainhole's negative far-end mass μ_- rises through zero at $v_e \approx 0.7$ and turns positive: fast rotation un-does the static drainhole's negative-mass end.

In sum, the backgrounds handed to evolution are verified along four independent lines: by identities the solver never imposes, through the full Einstein residual (Fig. 1); by closed-form exact solutions, extremal Kerr symbolically and the static members to round-off (Sec. II A); by the independent solver of Refs. [14, 15], matched to the precision a printed figure carries (Fig. 2); and by the evolution code itself, an independent discretization whose loaded data reproduce the solver's throat radii and global charges at $t = 0$ (Sec. IV).

IV. NUMERICAL SETUP

A. 3D initial data

The stationary metric (3) goes onto the Cartesian AMR grid through the isotropic radius r, $\eta = r - \eta_0^2/4r$: the far end $\eta \to -\infty$ compactifies to the puncture $r = 0$, as for the static drainhole [10]. The induced metric and the fields are read from the tables; the lapse and shift start from the stationary data, $\alpha = e^{f/2}$, $\beta^\varphi = -\omega$, so the extrinsic curvature $K_{ij} = (D_i\beta_j + D_j\beta_i)/2\alpha$ has only $r\varphi$ and $\theta\varphi$ components and the scalar momentum vanishes; at the puncture the frame rotates at ω_∞.

Three checks validate the pipeline. (i) At $t = 0$

TABLE II. The two families at representative spins, in units of η_0 ($G = c = 1$): rotation velocity v_e, angular momentum, mass of the $\eta \to +\infty$ end, ratio of end masses, scalar charge, equatorial throat radius, flattening, and throat angular velocity—each read off the solution's functions as defined in Sec. II. Every row is a converged solution that passed the acceptance gates of Sec. III. The published record covers family (b) only, as curves [14, 15], and we use it for validation; family (a), the one we evolve, is mapped here for the first time.

family	v_e	J/η_0^2	μ_+/η_0	μ_-/μ_+	D/η_0	R_e/η_0	R_p/R_e	$\Omega\,\eta_0$
(a) drainhole	0.213	0.2535	0.5574	−0.850	1.1004	2.0319	0.9873	0.1046
	0.513	1.0278	0.9302	−0.339	1.0007	2.6271	0.9194	0.1952
	0.699	3.0417	1.6557	−0.032	0.8598	3.8864	0.8351	0.1798
	0.900	34.550	5.8400	+0.179	0.5327	11.911	0.6926	0.0755
	0.950	152.16	12.316	+0.201	0.3785	24.751	0.6504	0.0384
	0.990	4053.8	63.666	+0.208	0.1698	127.36	0.6165	0.0078
(b) symmetric	0.239	0.0542	0.0252	1	0.9881	1.0299	0.9913	0.2323
	0.434	0.1158	0.0906	1	0.9583	1.1101	0.9699	0.3911
	0.667	0.2683	0.2666	1	0.8854	1.3420	0.9197	0.4969
	0.847	0.7007	0.6305	1	0.7653	1.8836	0.8440	0.4499
	0.898	1.0989	0.8685	1	0.7043	2.2722	0.8092	0.3952
	0.952	2.4701	1.4383	1	0.5971	3.2703	0.7539	0.2911
	0.990	12.235	3.4312	1	0.4157	7.0595	0.6795	0.1402

the Hamiltonian and momentum constraints converge at fourth order inside the throat region and are limited outside only by the tables' interpolation error (growing as $1/\Delta x^2$ and $1/\Delta x$ respectively), and the throat's ray radii converge to the solver's. We check this on the drainhole at $v_e = 0.51$ on three grids (ray radii within 5×10^{-5} at $N = 192$, $L = 64$) and, across the spin range, on the symmetric family at seven spins $v_e = 0.24$–0.99 with three resolutions each (within 7×10^{-5} at $v_e = 0.24$ to 9×10^{-4} at 0.99, $N = 256$). (ii) On the same symmetric set, the 3D ADM mass integral evaluated on the grid equals the same integral on the exact table metric to $\leq 10^{-12}$, and its Richardson extrapolation meets the solver's mass to 2×10^{-5}–7×10^{-4} (at worst 0.7% relative, at $v_e = 0.99$); the ADM angular momentum matches to between 2×10^{-3} ($v_e = 0.24$) and 2×10^{-5} (0.99). (iii) Dynamically, the static drainhole run through this pipeline is a numerical twin of the production hold of Ref. [11] over the full record: $t = 0$ bit-identical, Hamiltonian norm within 5×10^{-7} for all $t \leq 100$, and collapse milestones within one time unit.

B. Evolution scheme

Evolution uses `GRTeclyn` [26–28] with the setup of Refs. [10, 11]: CCZ4 [23], 1+log slicing [24], Gamma-driver shift [25], fourth-order stencils, RK4, Kreiss–Oliger dissipation, and a floored conformal factor at the puncture. The timestep is $\Delta t = 0.02\,\Delta x$ in every run. That value was adopted empirically in Refs. [10, 11]; we traced what set it to a grid-scale mode pinned to the fixed coarse–fine interfaces of the nested refinement boxes, dissipation-starved at $\sigma = 0.1$, whose growth rate rises with the timestep. Constraint norms evaluated per refinement level, with cells covered by finer levels masked, expose it by $t \approx 10$; the domain-wide norm is blind to it until breakout. Raising the dissipation where the mode lives removes it—at $\sigma = 0.4$ the interface-level norms sit 10^2–10^4 below the $\sigma = 0.1$ reference by $t = 100$—but the larger timesteps this would permit do not transfer to the spinning backgrounds, which lose the far side by $t \approx 22$–48 at $\Delta t = 0.035$–$0.05\,\Delta x$. The production runs therefore take no speed-up: $\Delta t = 0.02\,\Delta x$ everywhere, and the only scheme parameter that varies along the ladder is the dissipation, $\sigma = 0.1$ on the slow rungs and 0.4 on the fast ones (Table III). Its one measured side effect, a constant ∼4 time-unit advance of the collapse clock from the stronger initial smoothing, keeps rate comparisons at matched σ.

C. Long evolutions: the outer boundary

Evolutions that must outlive the outer-boundary crossing time $t \approx L$ need the boundary made harmless. Kreiss–Oliger dissipation cannot do it—its strength falls as the sixth power of wavelength, so the sponge shell of Refs. [10, 11] is transparent to the long outgoing gauge pulse—and a flat-space Sommerfeld condition assumes asymptotics the wormhole does not have (mass 1–12 sits inside the box). Two changes in the same outer shell fix it. The Sommerfeld condition is imposed on the deviation of the evolved fields from the stationary background rather than on the fields themselves; and a damping layer can in addition drive every evolved field back to that background at rate A, $\partial_t u \to \partial_t u - A\,f(r)\,(u - u_{\rm bg})$, with $f(r)$ the sponge's quartic ramp and $u_{\rm bg}$ the run's own

initial data, rebuilt exactly from the background tables; $A = 0$ recovers the base scheme identically. The layer is validated on the static hold at $A = 1$ against the reference evolution: $|K|$ falls two orders of magnitude across the shell, while everything inside it—the throat radii to four digits, the constraint norms, the near-zone ring—is unchanged, so it absorbs at all wavelengths without touching the interior. On the $L = 64$ box, whose shell ($r = 24$–32) contains the outermost extraction sphere, a side-by-side test showed the background-relative boundary alone keeps the outer levels cleaner, so those runs take the layer off; the $L = 128$ boxes ($r = 48$–64, outside every sphere) run with it on (Table III). With the boundary treated this way the $L = 64$ runs, previously trusted only to the crossing time $t \approx 64$, run to their stop times $t = 100$–150, with trust windows set by the constraints instead. The same comparison shows the late-time growth previously attributed to boundary reflection is sourced by the collapse region itself, and is removed by the raised dissipation of Sec. IV B.

D. The runs

Stability claims follow the triplet protocol of Ref. [22]: a quoted growth must have an e-fold time that converges to a nonzero value across resolutions, constraints that converge at the code's order while it grows, and a gauge-invariant measure—the throat's areal radius, or the marginally outer-trapped surface (MOTS) found spectrally on every output—growing with it. Every run carries three nested refinement levels on the throat, and the box is picked by the rung's mass: $M_{\rm ADM}$ triples along the family (1.11 at $v_e = 0.21$, 3.31 at 0.70), so the face of the $L = 64$ box recedes from $25\,M_{\rm ADM}$ of clearance at slow spin to under $10\,M_{\rm ADM}$ at $v_e = 0.70$. Rungs up to $v_e = 0.51$ therefore run on the $L = 64$, $N = 128$ grid of the static hold of Ref. [11] (finest $\Delta x = 0.0625$), and rungs from $v_e = 0.59$ on $L = 128$, $N = 256$ at the same finest Δx. Ψ_4 is extracted on spheres at $R = 10$–30 ($L = 64$) and $R = 14$–45 ($L = 128$); the signal is the $(2, 0)$ mode. Resolution triplets anchor the two ends of the ladder, $N = 96/128/192$ at $v_e = 0.21$ and $N = 128/192/256$ at $v_e = 0.90$ (Appendix B). Table III lists every run: nine spins from the static reference through the $O(J^2)$ regime [17, 18] to far beyond any perturbative result, with the $v_e = 0.51$–0.70 rungs being remeasured on the $L = 128$ box at this draft.

V. THE INSTABILITY ALONG THE LADDER

Perturbative initial data fail first. Our first spinning holds used the $O(J^2)$ backgrounds of Ref. [18] at $J = 0.25$ and $J = 1$ on the drainhole family. Both throats inflated where the static throat collapses—the same unstable mode with the opposite sign ($\tau = 5.8$ and 5.6). But expressed as the equivalent radial kick of Ref. [11], the push was $\varepsilon \approx -3 \times 10^{-5}$ at $J = 0.25$ and -7×10^{-3} at $J = 1$: growth by a factor $\sim$230 for a factor 4 in J, i.e. $\sim J^4$—the size of the $O(J^2)$ data's own truncation error, which the independent Einstein residual puts at 10^{-2} at $J = 1$ (Sec. III). Second-order data seed the very instability under study at their own error level; nothing at slow spin can be concluded from them.

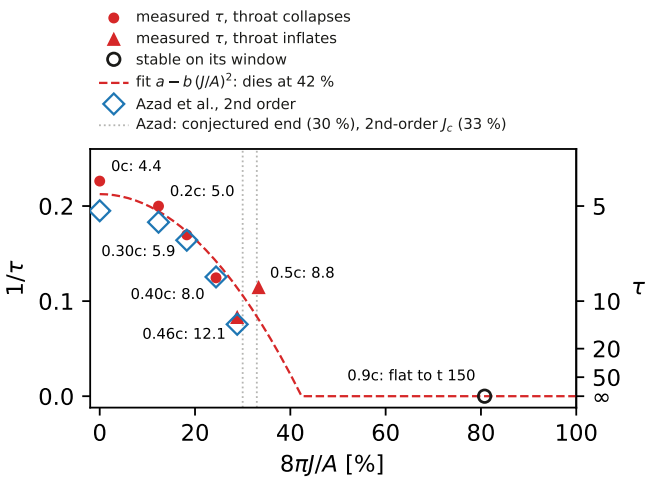


FIG. 4. One rate, two fates: the measured rate $1/\tau$ (left axis; τ right) against the angular momentum as a fraction of its extremal bound. Filled circles: collapsing rungs, τ from the log-linear fit of the areal-radius deficit over the 0.2–1% band. Filled triangles: inflating rungs, τ from $\ln|dR_A/dt|$ over the +0.2–1% growth band (the same fit on a pure exponential). Open circle: the 0.9c, flat to $t = 150$ at three resolutions. Dashed: the fit $a - b\,(J/A)^2$ through both branches, which reaches zero at 42%. Open diamonds: the second-order rates of Refs. [17, 18] at $C = 1$; dotted rules: their conjectured end of the instability ($\approx$30%) and the zero of their second-order rate (33%). The outcome: the mode slows with spin, changes fate from collapse to inflation near 30% instead of dying there, and is unmeasurably slow by 81%.

The collapse branch. On exact data (Table III) the verdict reverses, and the unperturbed throat collapses on the static mode, later the faster it spins: the areal deficit passes 1% at $t = 43.7$, 46.9, 51.4 and 60.2 along the rungs $v_e = 0$–0.40, with e-fold times $\tau = 4.4$, 5.0, 5.9, 8.0 (the log-linear fit of the areal-radius deficit over the 0.2–1% band, the one definition used throughout). Each collapse ends in a black hole: the 0.2c horizon settles at $R = 2.524$ ($M_{\rm MS} = 1.26$), the 0.30c at $R = 2.567$ by $t = 130$; the 0.40c's MOTS leaves the throat at $t = 92$ and reads $R = 3.22$, still shrinking, at $t = 100$. In each case the hold is seeded only by discretization error, so the ordering of rates is the ordering of $|\omega_I|$.

The inflating branch. At $v_e = 0.456$ (28.8% of extremal) the fate changes: the areal radius dips to -1.12% at $t = 83$, turns, crosses zero at $t = 101$ and grows exponentially, $\tau = 12.1$. The 0.5c behaves the same way (-0.30% at $t = 65$–68, then +6.9% by $t = 100$, $\tau = 8.8$), and the MOTS is lost once a throat expands—no trapped

TABLE III. The runs: the drainhole family (a) along the ladder, in code units ($\eta_0 = 2$; the static member has $M = 1$). $8\pi J/A$ is the angular momentum in per cent of its extremal bound and $M_{\rm ADM}$ the mass of the $\eta \to +\infty$ end (both from the background tables). Every run: three nested refinement levels on the throat, $\Delta t = 0.02\,\Delta x$; the finest Δx is 0.0625 at $N = 128$ on $L = 64$ and at $N = 256$ on $L = 128$, and scales inversely with N within a triplet. Recipe S (slow): $\sigma = 0.1$, $\chi \geq 10^{-8}$; F (fast): $\sigma = 0.4$, $\chi \geq 10^{-4}$. Shell: the damping layer of Sec. IV C on (✓) or the background-relative boundary alone (–). Trust: the last time the run is quoted. The last three rows are in flight at this draft.

rung	v_e	$8\pi J/A$	$M_{\rm ADM}$	L	N	recipe	shell	stop t	trust	outcome
0c	0	0	1.000	64	128	S	–	100	60	collapses, $\tau = 4.4$; dies inside its horizon, $t = 62.7$
0.2c	0.210	12.3	1.112	64	96/128/192	S	✓	100	55/100/100	collapses, $\tau = 5.0$; horizon settles at $R = 2.52$
0.2c, kicks	0.210	12.3	1.112	64	128	S	–	100	42/61	$\varepsilon = \pm 10^{-3}$: pushed out collapses, in inflates
0.2c, far zone	0.210	12.3	1.112	128	256	S	–	150	150	the $(2,0)$ burst on spheres $R = 14$–45 (Fig. 5)
0.30c	0.304	18.3	1.248	64	128	S	–	100, 130	130	collapses, $\tau = 5.9$; horizon at $R = 2.567$; full burst
0.40c	0.395	24.4	1.448	64	128	S	–	100	100	collapses, $\tau = 8.0$; MOTS leaves throat at $t = 92$
0.46c	0.456	28.8	1.634	64	128	S	–	100, 150	100, live	-1.1% at $t = 83$, inflates from $t = 101$, $\tau = 12.1$
0.5c	0.513	33.3	1.860	64	192	S	✓	100	100	shrinks, -1% at $t = 99$: the seed's other sign
0.5c	0.513	33.3	1.860	64	128	S	–	150	95	inflates from $t \approx 70$, $\tau = 8.8$
0.5c	0.513	33.3	1.860	128	256	F	✓	150	–	running
0.6c	0.592	40.3	2.305	128	256	F	✓	150	–	running
0.7c	0.699	51.4	3.311	128	256	F	✓	150	–	running
0.9c	0.900	80.8	11.68	128	128/192/256	F	✓	150	150	stable: areal drift $\leq 7 \times 10^{-5}$ at all three N

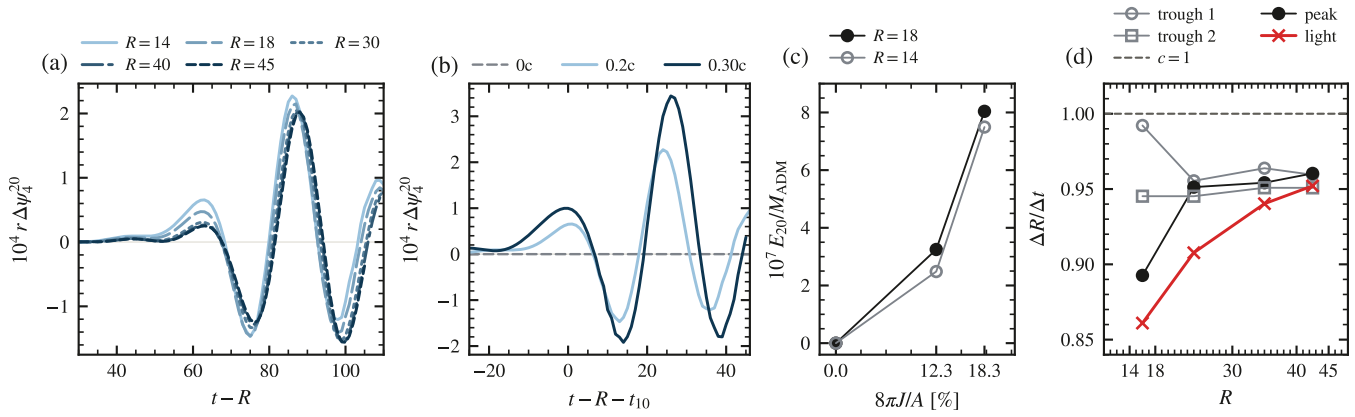


FIG. 5. The $(2,0)$ wave of the unperturbed rotating collapse. (a) $r\,\Delta\psi_4^{20}$ of the 0.2c far-zone run on spheres $R = 14$–45 against retarded time: one waveform with ring-down period ≈24; only the first lobe falls with R (near field), the extrema are flat in R to 1.5% over $R = 30$–45, and this box reproduces the $L = 64$ run's throat history to 3.2×10^{-4}. (b) The wave against spin at $R = 14$, each run on its own collapse clock $u = t - R - t_{10}$ (t_{10}: its equatorial deficit at 10%): the 0.30c burst has the 0.2c's shape and timing at ∼1.5× the amplitude, and the static run is drawn at zero—a spherical collapse radiates no quadrupole; its measured record stays below 3×10^{-8}. (c) The radiated energy $E_{20} = \frac{1}{16\pi}\int |\int r\psi_4\, dt'|^2\, dt$ over the aligned burst, per unit $M_{\rm ADM}$, at $R = 18$ (filled) and $R = 14$ (open): $3.3 \times 10^{-7} \to 8.0 \times 10^{-7}$ from the 0.2c to the 0.30c, a factor 2.5–3—the square of the amplitude ratio in (b)—with the static at zero. (d) The speed of each extremum between neighbouring spheres, against the local coordinate speed of light (crosses): the burst propagates at light speed. The outcome: an unperturbed, axisymmetric rotating collapse radiates a quadrupole burst whose energy grows with spin as the amplitude squared, and the static collapse radiates nothing.

surface forms. These are the two fates of one instability, and the seed's sign selects between them, exactly as declared kicks show at 0.2c (pushed in, the throat inflates; pushed out, it collapses; both pass 1% at $t \approx 25$ at the same rate): the 0.5c run whose absorbing shell gives the opposite effective seed shrinks instead (-1% at $t = 99$). At the fast end the 0.9c throat is simply stable, its areal radius flat to $\leq 7 \times 10^{-5}$ over the full $t = 150$ window at all three resolutions (Appendix B).

Against the perturbative prediction. Figure 4 puts both branches on one axis. The measured rate matches the second-order $C = 1$ rates of Refs. [17, 18] within 3% at $v_e = 0.30$ and 0.40 (5.9 vs 6.1, 8.0 vs 8.0); their expansion zeroes at 33% of the extremal angular momentum, and they conjecture the instability ends near 30%, where their main mode meets a second one. We find instead that the mode survives the crossing with its magnitude intact and only its fate changed: the 28.8% rung inflates with

$\tau = 12.1$ against their 13.2 (9%), and the 33% rung is still unstable where the second-order rate has already died. A quadratic fit through both branches reaches zero at 42%; the 0.9c, at 81%, is stable—answering for this family the four-dimensional stability question left open in Ref. [14]. The comparison is close but not like for like: their τ is a linear mode in Killing time, ours a nonlinear areal-radius fit. The $v_e = 0.51$–0.70 rungs are being remeasured on the $L = 128$ box at this draft, and the curve refills as they land.

Gravitational waves. The rotating collapse radiates with no nonspherical perturbation put in (Fig. 5): the 0.2c far-zone run carries one $(2, 0)$ burst from $R = 14$ to 45 at light speed, and the 0.30c repeats its shape and timing at $\sim$1.5$\times$ the amplitude, so the radiated energy per unit mass grows by the amplitude ratio squared, $3.3\times 10^{-7} \to 8.0 \times 10^{-7}$. The static collapse is spherical and radiates nothing—its record stays at the numerical floor. Rotation thus imprints itself on the burst a collapsing throat emits, before any horizon forms.

VI. CONCLUSIONS

Three results stand. First, the spinning Ellis–Bronnikov parameter space is now mapped as data, not curves: both families, from slow rotation to $v_e \approx 0.99$, as solutions exact to the 10^{-10} level at slow spin and the 10^{-7} level near extremality (Table II), including the massive drainhole family for which nothing was published—and including the correction of a field equation, Eq. (12) of Ref. [15], that closed-form extremal Kerr convicts directly. The practical lesson generalizes: a solver gated only on the equations it imposes would have passed every test while solving the wrong system; the unimposed identities caught it. Second, the pipeline from these solutions to 3+1 evolution is validated at fourth order, with the static limit a numerical twin of the production run of Ref. [11], and $O(J^2)$ initial data are shown to be unusable for stability work: their truncation error seeds the instability itself. Third, the first nonlinear evolutions of rotating wormholes show one rate with two fates: the radial collapse slows with spin ($\tau = 4.4 \to 8.0$ from the static to 24% of the extremal angular momentum), the sign predicted at $O(J^2)$ [17, 18]; near 30%, where the perturbative mode crossing was conjectured to end the instability, the unperturbed throat inflates instead, with the mode's magnitude intact; the measured rate extrapolates to zero at $\approx$42%, and at 81% the throat is stable to $t = 150$ across three resolutions—answering, for this family, the four-dimensional stability question left open in Ref. [14]. The rotating collapse also radiates a $(2, 0)$ burst from unperturbed, axisymmetric data, its energy growing with spin as the amplitude squared, while the static collapse radiates nothing. What remains on the ladder is in flight: the $v_e = 0.51$–0.70 rungs on the large box, which fix where between 33% and 81% the instability actually dies—and whether it dies monotonically or as the oscillating complex pair of the charged analogue [20]—and the end states of the inflating branch.

ACKNOWLEDGMENTS

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Appendix A: Solver details

The unknowns f, ω are expanded in Chebyshev polynomials of $x = (2/\pi)\arctan(\eta/\eta_0) \in [-1, 1]$, which compactifies both ends, and in even cosine modes of θ; derivatives are spectral, and the nonlinear system (4)–(5) is solved by Newton iteration with a direct linear solve. At each Newton step, ν and the constant D^2 are recovered together from the overdetermined pair—the first-order constraint (6) and the D^2 identity—as one linear least-squares problem, which is what makes the acceptance gates of Sec. III meaningful: the reported D^2 spread, Smarr defect, flux equality and full-Einstein residual are all computed from equations that never entered the solve. Spin is walked with J held fixed in each solve: ω_∞ joins the unknowns and one extra row imposes the target J, since ω_∞ is not monotone along a family (Table I). Steps are taken in $\log J$, predicted by the secant through the last two solutions, growing after fast Newton convergence and halving after slow convergence or a failure; grids refine automatically from 48×12 when the acceptance gates demand it, to 162×42 at $v_e = 0.95$ on the drainhole family. On the symmetric family the mirror identities are satisfied at the 10^{-14} level, including $\omega(0) = \omega_\infty/2$, and the equatorial ergosurface law of Ref. [15] holds to 10^{-12} wherever the ergosurface sits at $\eta > 0$ ($v_e \geq 0.77$). The extremal-Kerr substitution of Sec. II A is kept in the pipeline as a symbolic regression test.

Appendix B: Convergence of the evolutions

Resolution triplets anchor the two ends of the ladder: the 0.2c, the slowest spinning collapse (Fig. 6), and the 0.9c, the stable fast rung (Fig. 7). In both, the constraint norms are evaluated per refinement level with cells covered by finer levels masked (Sec. IV B), and the coarsest member's norms are compared against the finest member's rescaled by 2^2 and 2^4—where they would sit at second and at fourth order.

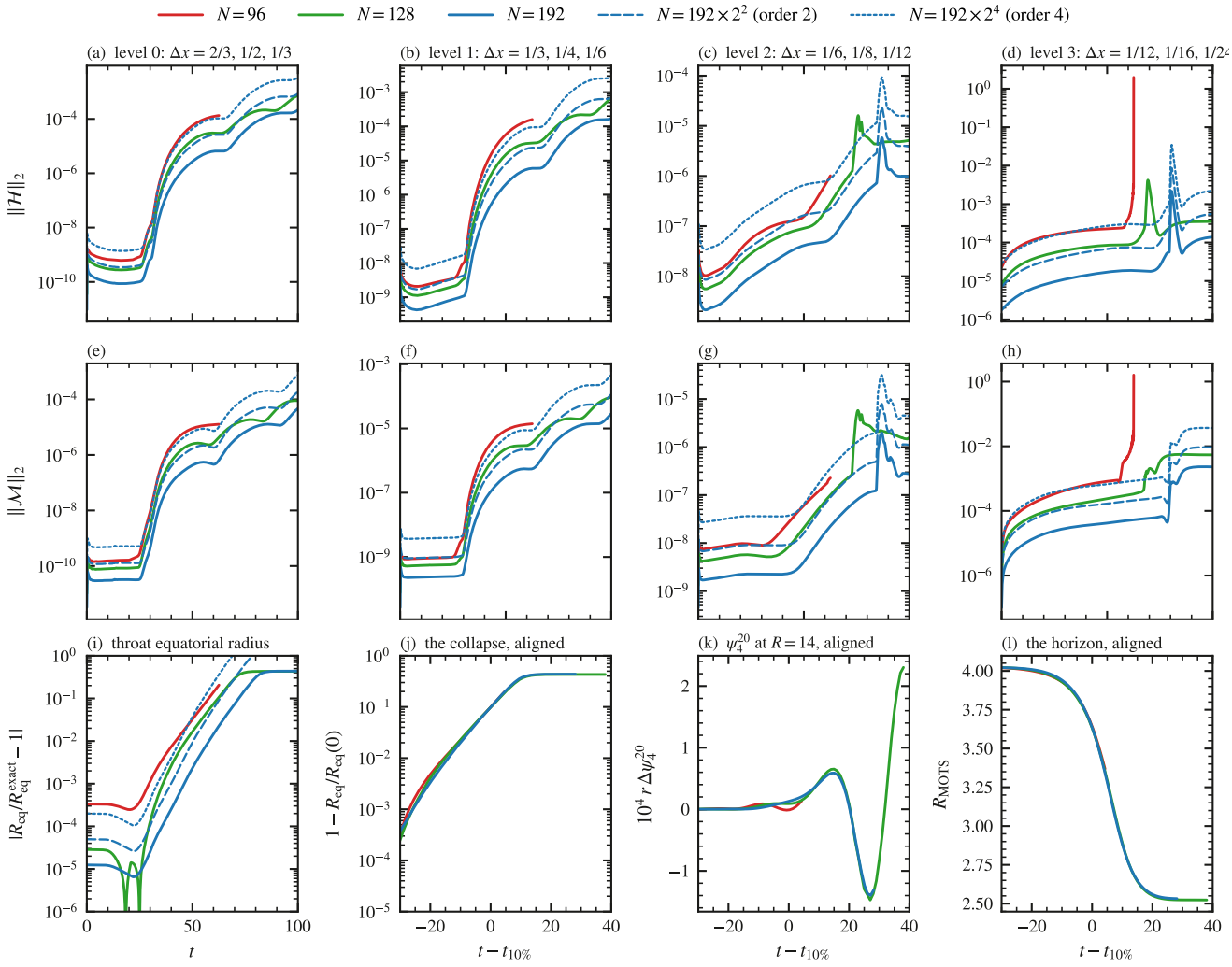


FIG. 6. Convergence of the 0.2c rung over its triplet $N = 96/128/192$ ($L = 64$; finest Δx in each panel title). (a)–(d) Hamiltonian and (e)–(h) momentum constraint norms per refinement level; dashed and dotted, the $N = 192$ norms rescaled by 2^2 and 2^4: the measured $N = 96$ norms lie between the second- and fourth-order brackets on every level while the collapse grows. (i) The equatorial throat radius against the stationary solution, falling with resolution. (j) The collapse, each run aligned at its own 10% time: the three resolutions land on one curve. (k) $r\,\Delta\psi_4^{20}$ at $R = 14$, aligned: $N = 128$ and 192 overlap ($N = 96$ dies in its collapse before the burst arrives). (l) The MOTS radius, aligned: one horizon, settling at $R = 2.52$. The outcome: the collapse rate, the waveform and the horizon are grid-converged at third to fourth order.

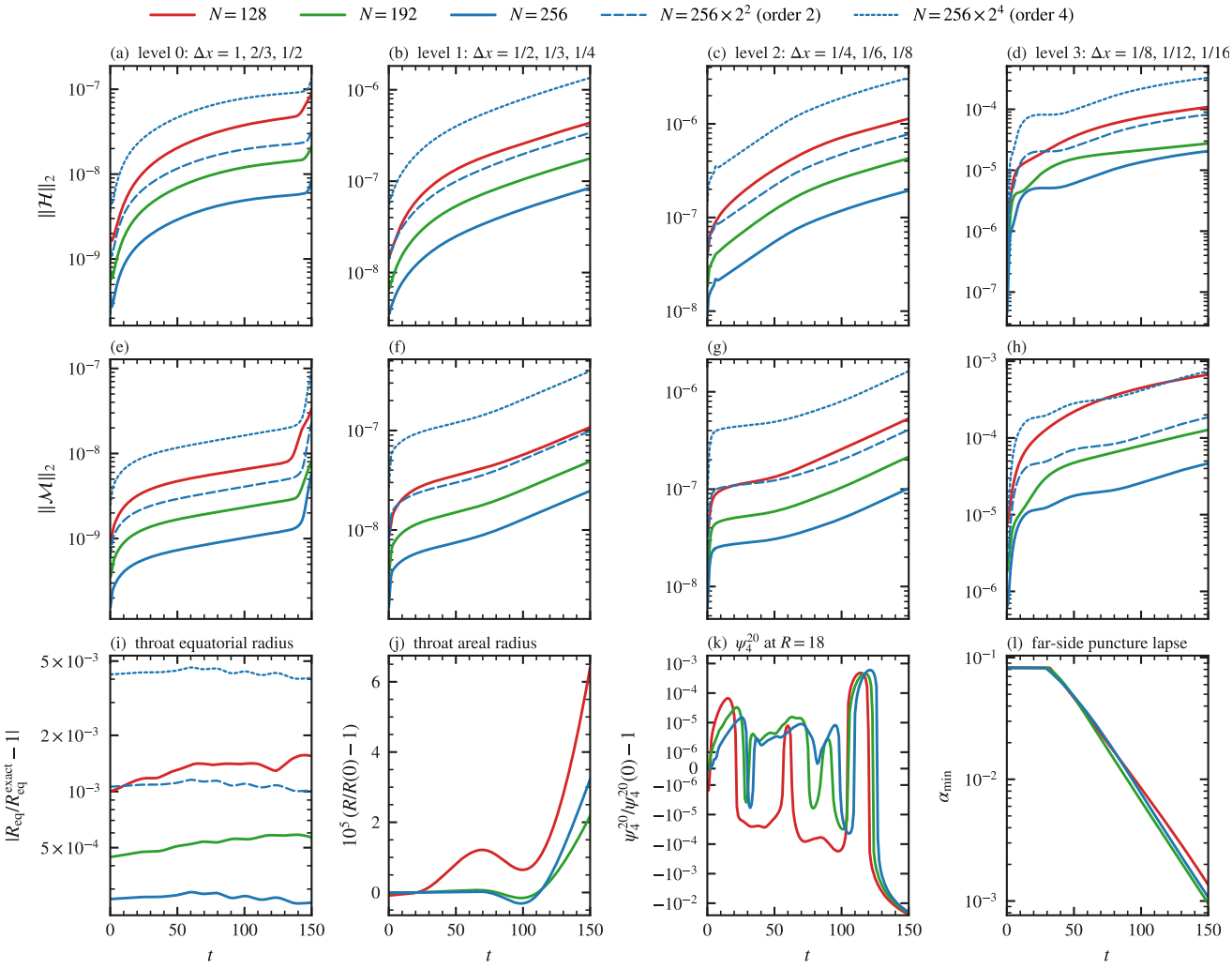


FIG. 7. Stability of the 0.9c rung over its triplet $N = 128/192/256$ ($L = 128$; finest Δx in each panel title), to $t = 150$. (a)–(h) Constraint norms per level with the same second-/fourth-order brackets as Fig. 6: every level converges with resolution and nothing breaks out over the window. (i) The equatorial throat radius holds the stationary value, its offset falling with resolution. (j) The areal throat radius drifts by $\leq 7 \times 10^{-5}$ over the full window. (k) The $(2, 0)$ extraction at $R = 18$, relative to its $t = 0$ read-off: fluctuations at the 10^{-4} level, no burst. (l) The far-side puncture lapse falls smoothly at all three resolutions (the gauge, not the geometry). The outcome: the 0.9c throat is stable on its window at every resolution—no growing radius, no constraint breakout, no wave.

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